Introducing Quail: An Environment for Agentic Qualitative Analysis

Presenters

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Abstract

While text retrieval shows an agent a few examples, Quail gives it a Python kernel and an analysis language, in order to make sense of the whole dataset. Researchers often have more text than anyone can read closely [1]. LLMs can help, and one common way is through retrieval-augmented generation (RAG) [2], which gives a language model the few entries that best match a query. Standard RAG can find what was said about a topic but struggles with questions that require context over the whole dataset, like how often something was said and how it changed over time.

Notebooks like Jupyter let researchers iteratively explore. Quail gives an agent the same loop, where each cell (a block of Python) from the agent runs in a kernel that stays alive between cells. After each one, the agent receives printed output and the last line's value, just what a notebook would show. Many systems give an agent a separate tool for each operation. Quail instead relies on an analysis language inside its Python kernel, where functions to search, count, and tag are highly composable. Agents often work better this way than by calling tools one at a time [3], an approach now commonly called "code mode" [4].

Inside the Python kernel, Quail's analysis language acts as the bridge between the kernel and the dataset. The components of the language build on one another, from a dataset's columns to the four core functions. The language also gives a researcher and an agent a concise, shared vocabulary; this allows a researcher to efficiently interrogate an agent's actions.

Sessions in Quail are organized into studies. Because a study is a folder of text files, git can share it the way any piece of code is shared. The file structure is designed so that agents working in parallel can push logs without conflict and pick up where another agent left off. Semantic search needs the entire search space to be embedded, which can take hours to compute. `quail warm` can do this work ahead of time, distributed on several machines.

Operating Quail requires only a shell, so it can run wherever an agent does, on a laptop or a cloud VM. In one tested setup, TV news transcripts from six U.S. presidential election years, a Carleton College research dataset, were prepared as a study. Mentioning @Cursor in a Slack thread started a new Cursor cloud agent, whose VM booted from a snapshot that already held Quail, an embedding model, and the election news transcripts, indexed and embedded. Asked how Obama is characterized, the agent ran nearly 20 cells and replied 27 minutes later. Quail 0.94 is an early release for Linux and macOS, open source under the Apache-2.0 license [5].

References

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Connection to Mission, Goals, & Interests of US-RSE Community

This poster connects to US-RSE's Resources and Advocacy goals by presenting Quail as a reference for RSEs building responsible AI-assisted research software. Quail is designed to be discussed and built upon in public with its Apache 2.0 License. In the context of this year's conference, USRSE'26, Quail takes a step forward in qualitative research science by offering a case for utilizing generative AI in certain research settings.

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